Science and Justice 52 (2012) 145–160

Contents lists available at SciVerse ScienceDirect

Science and Justice

journal homepage: www.elsevier.com/locate/scijus

A methodology for finger mark research

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ARTICLE INFO

Article history:
Received 23 September 2011
Received in revised form 22 October 2011
Accepted 24 October 2011

Keywords:
Finger mark
Experiment
Methodology
Research
Enhancement

ABSTRACT

Currently there is no standard way of carrying out research into finger mark enhancement techniques. Individuals, groups or establishments tend to use different methodologies depending on a number of factors, especially finance and time. However, data published in the literature can be misleading to the forensic community if the data generated reflects research involving very few finger marks or if those finger marks have been deliberately doped with an unnatural balance of sweat or an unusual contaminant.

This paper presents an experimental methodology which is intended to establish minimum standards for those carrying out finger mark enhancement research (at least within the United Kingdom) and bring some consistency to the process. It will aim to identify the many variables encountered when dealing with finger marks and suggest experimental methods to take these into account. It will also present the key stages of the progression of a process from a laboratory concept to a tool used on operational work.

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1. Introduction

Finger marks have been used as identification evidence for over 100 years. Early criminal cases where finger mark evidence was presented typically utilised finger marks that were already visible ('patent' marks) due to them being deposited in substances such as blood, grease or paint. However, since those early cases the number of finger marks recovered from crime scenes has been increased significantly by the discovery of techniques capable of enhancing invisible ('latent') marks so that they can be located and imaged. The earliest of these enhancement techniques, iodine, was noted as early as the 1860s [1] and during the subsequent 150 years several hundreds of different techniques have been proposed for finger mark enhancement.

These techniques may operate in many different ways, some reacting with specific chemical constituents of finger marks (e.g. amino acids, lipids, chlorides), others utilising some difference in property between the finger mark and the substrate it has been deposited on, and others utilising other properties (e.g. adhesive properties, optical properties, fluorescence) of the mark to differentiate it from its background. Because the approaches used to enhance finger marks are so different, it is necessary to have some standardised means of comparing the relative performance of enhancement techniques. This enables the techniques most appropriate for operational use to be identified, facilitating informed decisions about which techniques to use for a particular scenario, and which order they should be used in.

A standardised methodology is also important when exploring novel processes for finger mark enhancement, which may show promise in initial experiments but require further testing to establish if they are suitable for operational implementation. There is currently no widely published protocol for carrying out validation of techniques, although various research organisations may have local procedures. The publication of such a protocol would benefit both operational practitioners, ensuring that the techniques available for use are both rigorously tested and effective, and the research community, ensuring that published papers on enhancement techniques are consistent in the way results are obtained and compared.

This paper aims to present such a methodology, describing the approach used in the finger mark research conducted by the United Kingdom Home Office. This approach has been used for approximately 40 years, and has provided the data underpinning the advice given in the Manual of Fingerprint Development Techniques [2]. Elements of this methodology have previously been outlined by Kent [3], and are covered in greater depth in this paper.

The fundamental issue that needs to be addressed in any assessment of a finger mark enhancement technique is the variability of finger marks, both between the marks deposited by different people and between marks deposited by the same person over a period of time. If this variability is not taken into account in experiments, then a false impression of the effectiveness of the technique may be created.

The fingers and palm contain exclusively eccrine glands [4,5] so it may be expected that finger marks would contain only eccrine sweat. However this is not the case. In practice fingers make regular contact with other parts of the body and may therefore pick up sweat secreted by the sebaceous and apocrine glands [4,5]. In addition, fingers can also pick up contaminants from any other surface that is touched. Once these substances combine on the fingertip, there is potential for interaction and reaction between all the chemicals present. The amount of sweat material produced may vary considerably

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doi:10.1016/j.scijus.2011.10.006

depending on factors such as the environmental conditions and psychological moods of the person. An example of the variability in finger marks deposited by the same donor over a 4 h period is illustrated in Fig. 1.

After deposition the finger mark can undergo further alterations [6] as chemicals within the complex mixture continue to react and interact, not only with each other, but also with the surface on which the finger mark has been deposited. These compositional alterations may, in some cases, be directed by environmental factors such as temperature, humidity, air flow and light.

It can thus be seen that each finger mark represents an individual and complex mixture of secretions from the body and contamination from the local environment that has physical as well as chemical structure. There are many other variables that can have an effect on the chemical composition of a finger mark. An overview of some of the most important factors is given in Table 1.

The complexity of finger mark composition described above means that for any single finger mark there may be many techniques that are capable of enhancing it, either by interaction with the chemicals in it or some other property.

Ideally, experiments in finger mark enhancement should follow basic scientific method (an experiment, within this context, is a study of cause and effect and it differs from non-experimental methods in that it involves the deliberate manipulation of one variable, whilst maintaining consistency in all other variables). However, as outlined above there are many variables relating to finger mark deposition and most of these cannot be controlled so that they are consistent across a range of individual experiments. As a consequence, a viable means of conducting finger mark development research is to use significant numbers of marks to establish clear trends, whilst making reasonable attempts to keep control the relevant variables (such as contaminants) and maintaining environmental factors within pre-defined limits.

An essential feature of the test methodology outlined in the following sections is that the largest part of the test programme utilises 'natural' finger marks collected from donors conducting their normal daily routine, as opposed to 'grooming' of finger marks (i.e. where donors wipe their fingers across the face or forehead prior to deposition of marks). Experimental studies [7,8] have shown that this practice of 'grooming' significantly changes the composition of the finger marks deposited, usually raising the lipid content and potentially biassing results in favour of any technique targeting these constituents. Groomed finger marks are also less representative of the types of marks that are encountered in operational work. This means that any bias in experimental results is potentially compounded if the technique is introduced into operational use because it is being applied to finger marks that typically contain appreciably less lipid material than those used in laboratory trials.

Table 1
Some of the factors influencing the nature of a deposited finger mark.

Variable	Examples affecting finger mark
Glandular	Eccrine (fingers and palm only have eccrine glands); sebaceous; apocrine
Contaminants	Anything touched (as touching is a two-way process); pollution
Ambient conditions	Temperature; humidity; air flow; light
Donor factors	Time of day/year; diet; medication and recreational drugs; gender; age; mental and physical states; racial origin; cleaning regime
Deposition	Pressure; angle; movement at the moment of touch
Surface	Porosity (porous to non-porous ratio); condition; surface chemistry/physics, contamination; temperature
Biological	Bacteria; viruses

The four principal stages that may be used in the test methodology are as follows:

1. Finger mark constituents/test strips/groomed finger marks
2. Planted finger marks/laboratory trials
3. Pseudo-operational trials
4. Full operational trials

The purpose of each of these stages and some of the techniques that can be used will be described in greater detail in the sections below.

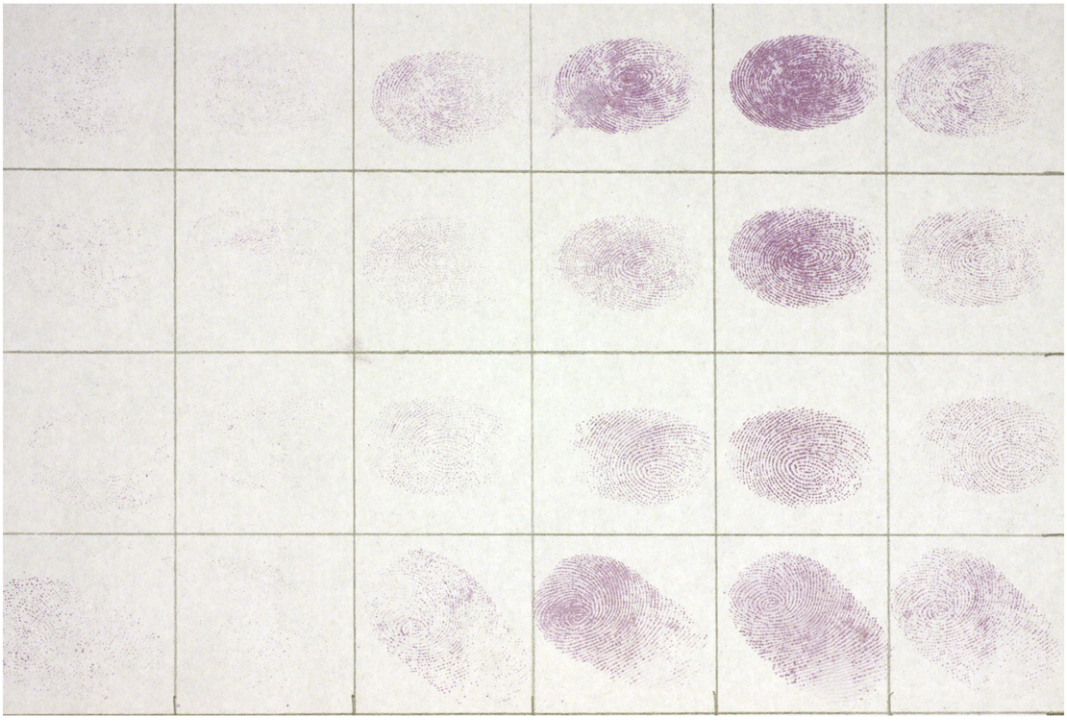


Fig. 1. A series of marks deposited by different fingers of the same donor at different times after washing hands with soap and water. Marks subsequently developed using ninhydrin. From left to right, time of deposition = 0, 30, 60, 120, 180, 240 min after washing hands, no further washing during deposition period.

2. Methodology

2.1. Phase 1. Finger mark constituents/test strips/groomed finger marks

This initial phase of testing may be carried out for several purposes:

1. To establish whether the proposed reagent or technique interacts with finger mark constituents (or properties) it is proposed to interact with, and to establish the outcome of this interaction.
2. To determine what the visual appearance and emission/absorption spectra of any reaction products might be.
3. To investigate the relative sensitivity of different formulations of a single reagent on a coarse scale as a pre-screening exercise.
4. To investigate the effect of changing elements of the formulation to optimise it for finger mark development.

Some of the methods that may be used in obtaining this data include; finger mark constituents, test strips and groomed or doped finger marks. Although the use of simulants such as 'standard' finger mark mixtures can speed up initial assessment of a technique, use of these cannot substitute for real finger marks beyond this initial stage of study. This is because a finger mark is much more complicated and behaves in a more intricate manner than a simple mixture of chemicals.

Further details of the types of study that may be performed in this phase are outlined below.

2.1.1. Establishing interactions between the finger mark and the technique

The objective of such tests is to establish what type of finger mark constituents (eccrine or sebaceous) the enhancement process is most likely to interact with (if this is unknown), and how strong this interaction is likely to be. In this initial trial, deliberately 'groomed' eccrine and sebaceous marks can be used. Marks are deposited by a small pool of donors and the enhancement technique applied to them. The number and quality of the developed marks are then recorded. Examples of different enhancement processes applied to 'natural', 'eccrine' and 'sebaceous' marks are illustrated in Fig. 2, clearly showing which types of constituents are targeted by particular techniques.

The ways in which loaded eccrine and sebaceous marks are generated are outlined below.

2.1.1.1. Generating eccrine finger marks. Subjects should wash their hands thoroughly with soap and water, making sure all soap residues are washed away. Then place the hand in a clean polyethylene bag for 20–30 min before donating finger marks. Note that clean, powder-free nitryl gloves may be used as an alternative to the bag, and longer times may be required to generate sufficient eccrine sweat at colder times of the year. (In the UK, interior temperatures in non-air conditioned rooms may be 15–20 °C during winter, but may exceed 30 °C and become more humid in summer months.)

2.1.1.2. Generating sebaceous marks. Subjects should wash their hands thoroughly with soap and water, making sure all soap residues are rinsed away. Fingers should then be dried and then rubbed on the nose before deposition. The forehead, whilst being a source of sebaceous sweat, also has eccrine glands making it less suitable. Note that if cosmetics have been applied to these regions these contaminants may dominate the deposit.

For processes that rely on chemical interactions between the finger mark residues and the enhancement reagent, it may be of additional interest to determine which specific constituents within those covered by the broad descriptions of 'eccrine' or 'sebaceous' the reagent is actually targeting. In some cases it may already be known what type of constituents (e.g. amino acids) the developer is intended for but it may still be necessary to either validate that the expected reactions are occurring with model constituents, or to assess how consistent this reaction is across a range of constituents (see Section 2).

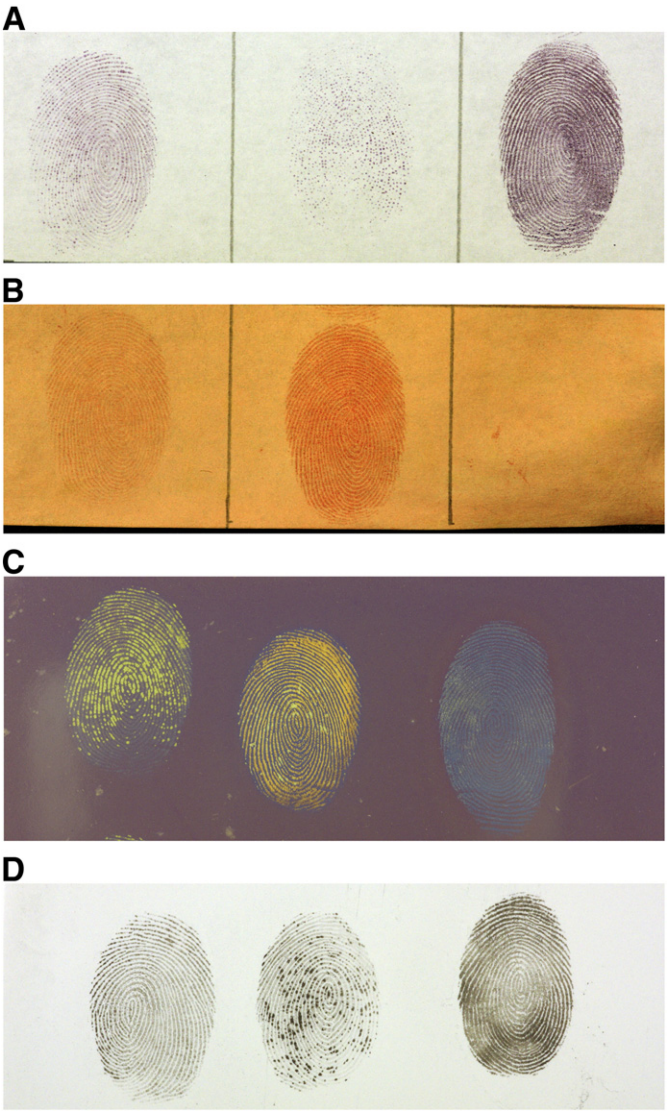


Fig. 2. A comparison of enhancement processes targeting different constituents. From left to right: 'natural', sebaceous and eccrine marks. From top to bottom: ninhydrin, Oil Red O, silver vacuum metal deposition, black powder suspensions.

2.1.2. Investigating characteristics of reaction products

One method for investigating characteristics of reaction products is to use solutions of model constituents that are deposited onto representative substrates (e.g. glass for non-porous surfaces, paper for porous surfaces). The enhancement technique is then applied to the surface once solutions of the model constituents have dried. The appearance of the reacted test spots can then be assessed to determine the colour and/or fluorescence of any reaction products. Sets of model compounds (amino acid kits, mixed lipids, salts) are commercially available and can be utilised for this purpose. Amino acid reagents lend themselves to this type of assessment because there can be a reasonable degree of variability between the reaction products formed with different amino acids and with other constituents of eccrine sweat, Fig. 3.

This use of test spots or squares is particularly useful when the reaction products are fluorescent, because the reacted paper samples can be placed into instruments such as a spectrofluorophotometer, and the excitation and emission spectra for the reaction products measured directly [9,10], Fig. 4. Filter paper is a suitable substrate for testing fluorescent reaction products on porous surfaces because

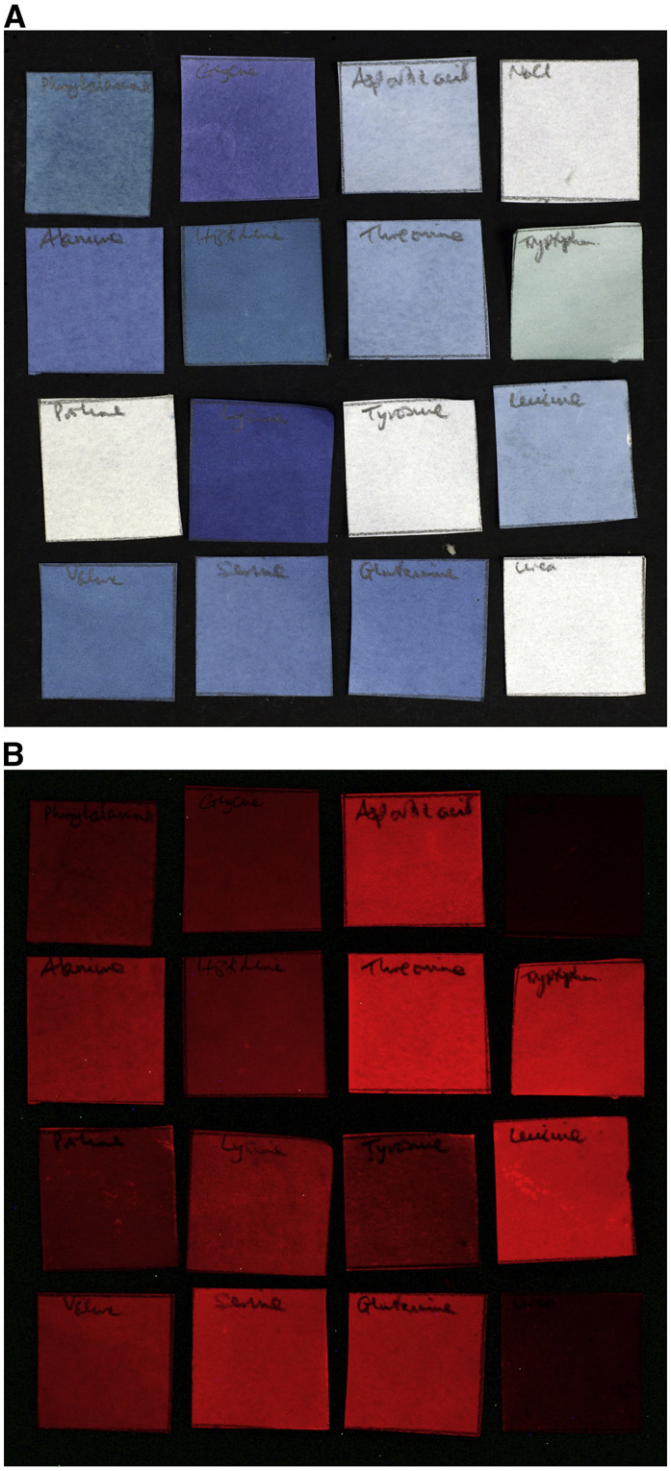


Fig. 3. Reaction products formed between genipin and ~0.1 M solutions of different eccrine constituents (amino acids, urea, sodium chloride) on filter paper, A) visual, B) fluorescent.

it is free of fillers and optical brighteners that would otherwise interfere with the fluorescence of the reaction product. Alternatively, the reaction can be carried out in solution and these spectra obtained from vials of liquid.

This information is useful in determining which light sources are most appropriate for illuminating articles, and in establishing the most appropriate filters for viewing the developed marks and photographing the fluorescence.

In some cases it may also be desirable to establish the pathways of the reactions that are occurring and the nature of any reaction

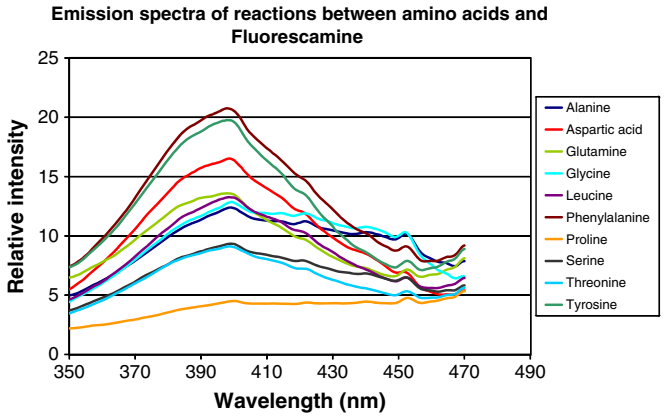


Fig. 4. Emission spectra for the reaction products formed between different amino acids and Fluorescamine.

products formed, both the final products and any intermediates. In some cases the expected reaction may not always proceed to completion and the intermediate reaction products may be what are encountered in practice. Studies of this type are well-suited to academic institutions with specialist equipment such as nuclear magnetic resonance spectroscopy (NMR), X-ray crystallography and gas chromatography mass spectrometry (GC–MS). Published studies by Wilkinson [11,12] illustrate how such techniques have been used to determine the nature of the reaction products formed between amino acids, DFO and 1,2 indandione.

2.1.3. Establishing technique sensitivity

During validation of a technique, it may also be necessary to extend the studies outlined above to look at known dilutions of model constituents. This has the objective of demonstrating that not only does the reagent react with the constituents it is expected to, but also that it is sufficiently sensitive to detect and react with the low concentrations of these constituents that may be present in real finger marks. For this type of test, 'dilution series' can be used, where spots of a solution of the model constituent (or known contaminant) are deposited at different concentrations, exposed to the reagent and developed, Fig. 5. The lowest concentration for which a visible and/or fluorescent reaction product is still formed can be recorded and compared to results from other reagents and/or variations on the same reagent [13,14].

Standardised 'test strips' containing mixtures of finger mark constituents printed at different concentrations may also be used for this purpose [15], Fig. 6.

2.1.4. Optimisation of formulation and/or processing conditions

Once a candidate formulation has been identified, it can be 'fine tuned' by assessing the effect of changes in some of the key components on the visible and/or fluorescent reaction products. One test that has been devised for this purpose is the 'quartered finger mark', where several different donors place a single natural finger mark in the centre of a surface which is divided into four equal parts, and a further finger mark deposited across each of the boundaries between quarters. The surface is then divided into four and each of these quarters is treated with a formulation differing in only the concentration of a single component from its neighbours. After treatment the quarters are recombined for assessment, looking at the intensity of the colour and/or fluorescence in each quarter across the range of donors used, Fig. 7. In this way the tolerance of the formulation to changes in concentration of the principal components can be established.

The 'quartered finger mark' approach can also be applied to testing processing conditions and times, for example using the same formulation on each quarter but processing each at a different temperature, or

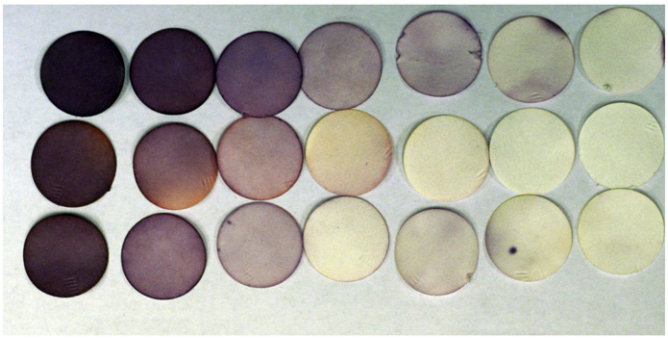


Fig. 5. Dilution series for three amino acids (serine, alanine and threonine) reacted with ninhydrin (1 mg mL^{-1}, 0.33 mg mL^{-1}, 0.11 mg mL^{-1}, 36.6 ng mL^{-1}, 12.2 ng mL^{-1}, 4.07 ng mL^{-1}, 1.36 ng mL^{-1}).

at the same temperature for different times. However, it must be ensured that each finger mark is as compositionally similar as possible by carefully rubbing fingertips together and that several donors are used so as to have as broad a range of finger marks as possible This method is especially effective where quick, clear objective measurements can be made, e.g. the effect of component concentration on the intensity of fluorescence produced.

The objective of all of the above experimental methods is to produce an optimised formulation that can be used in comparative trials with existing techniques. Only the tests most appropriate to the type of enhancement technique being evaluated need be utilised. This phase is not always required, for example when new powders are being introduced the 'formulation' is already set and this phase of testing is essentially bypassed.

2.2. Phase 2. Depletion series/laboratory trials

The concept of a 'Triangle of Interaction' between the finger mark composition, the surface and the environment are now being taught in the United Kingdom as an aid to decision making, Fig. 8. This concept suggests that by obtaining as much information as possible about

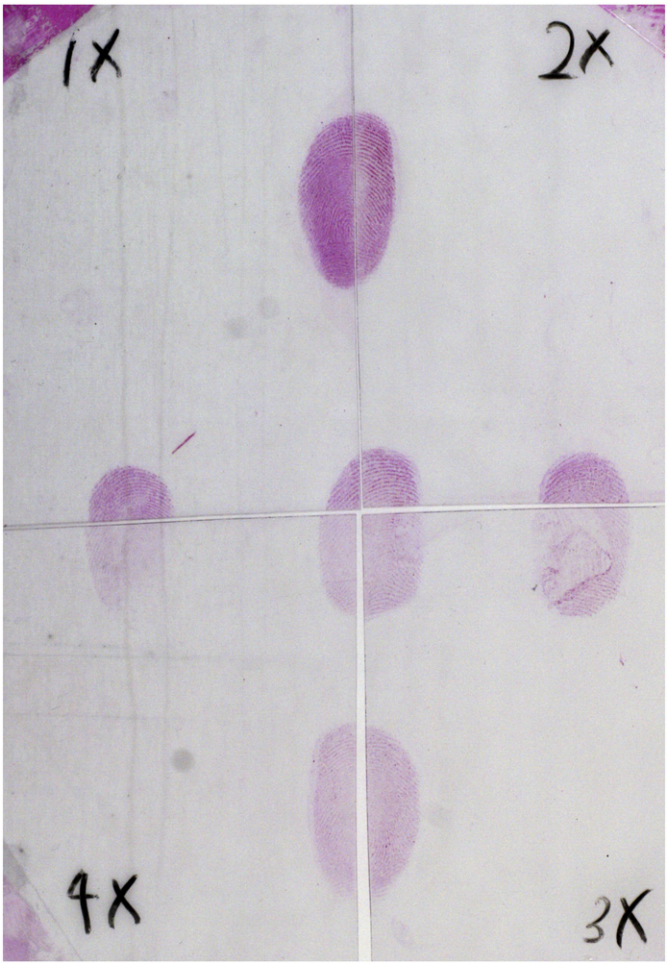


Fig. 7. A 'quartered finger mark' experiment investigating the effect of reducing the dye content in a formulation based on basic violet 2. 1× = original concentration, 2× = ½ original concentration, 3× = 1/3 original concentration, 4× = ¼ original concentration.

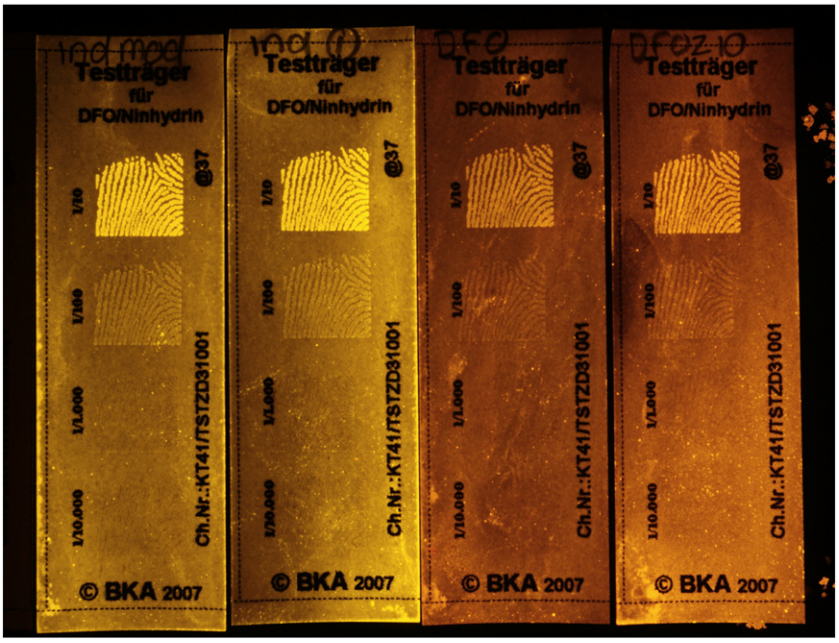


Fig. 6. Test strips printed with concentration gradients of amino acid mixtures and reacted with different formulations of amino acid reagents.

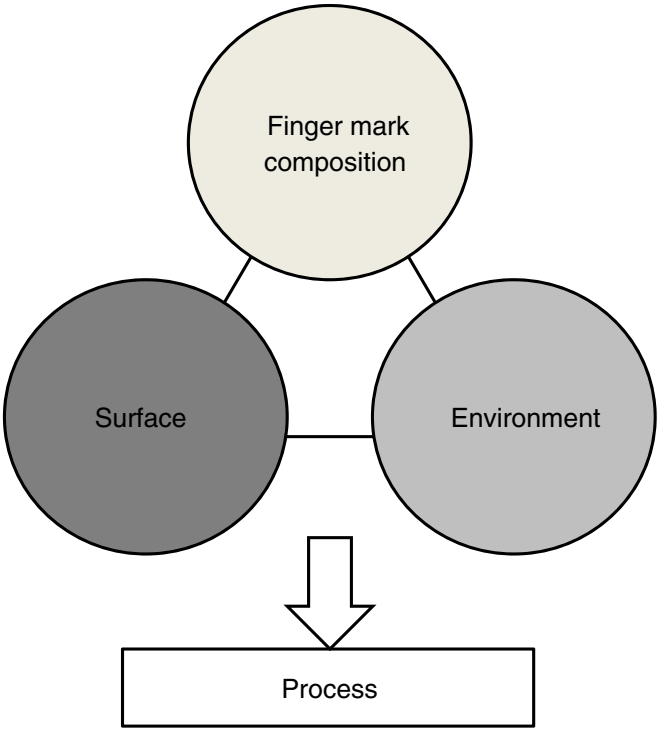


Fig. 8. The concept of the Triangle of Interaction.

these three factors and the interactions that may occur between them, better decisions can be made about the most appropriate enhancement process to use for a particular scenario.

This phase of testing essentially reverses the thought process of the 'Triangle of Interaction' by establishing the robustness of a development process to changes in finger mark composition (i.e. range of donors), surface type, and the environment (including age of mark).

Testing in this phase of the work should be the most extensive, using deliberately deposited finger marks throughout. These should be 'natural' finger marks, deposited by donors who have not washed their hands in the past 30 min, have not applied cosmetics in the same time period, and have not deliberately wiped their hands across regions of the face rich in sebaceous products. Prior to deposition the donors are asked to rub their hands together to try to achieve as even as possible a distribution of the finger mark residues across all fingers.

The objective of this phase is to carry out as direct a comparison as possible between the effectiveness of the development technique under evaluation and that of the existing technique(s) for that application.

When devising experiments for this phase it is important to keep them of manageable size and not try to compare every variable in one experiment. It is preferable to use a series of experiments exploring fewer variables and use these to build the overall picture. If at any stage one of the variations or techniques is not performing effectively its continued involvement in experimentation should be reviewed.

During such comparative studies, it is desirable to obtain information about the breadth of effectiveness for the technique (i.e. how many people in a typical population it will develop marks for) and the sensitivity of the technique (i.e. how small a trace of residue can still give positive development with the technique). To some extent the sensitivity aspect may have been explored in Phase 1 of the study, but this part of the evaluation process extends this to real finger marks.

To ensure that any conclusions drawn from this stage of the evaluation are as robust as possible, the comparison process should take into account the range of surfaces the technique may be applied to on operational work, the different ages of mark that may be encountered, and the variation in finger mark composition between donors. Some of the ways in which these factors can be addressed in laboratory trials are described below.

2.2.1. Multiple donor study

A multiple donor study is a means of establishing how many people in a sample population the technique will typically develop finger marks for. In this case, a single finger mark is taken from as many different people as possible (preferably >40), attempting to obtain a good mixture between ages, sexes, ethnicities, office workers vs. manual workers etc. The series of deposited marks are then aged for the appropriate period of time and then developed using the technique under evaluation, Fig. 9. The number and quality of the developed marks are then recorded. The results obtained will be specific to the surface type and age of mark used, and multiple tests of this type may be required to build an overview of performance.

This type of study may use the first, heaviest mark deposited by each person, or alternatively it may be possible to simultaneously obtain some information about sensitivity by using a mark deposited from the same finger after it has already touched a similar surface several times. The concept of 'depletion' is described in more detail in the next section.

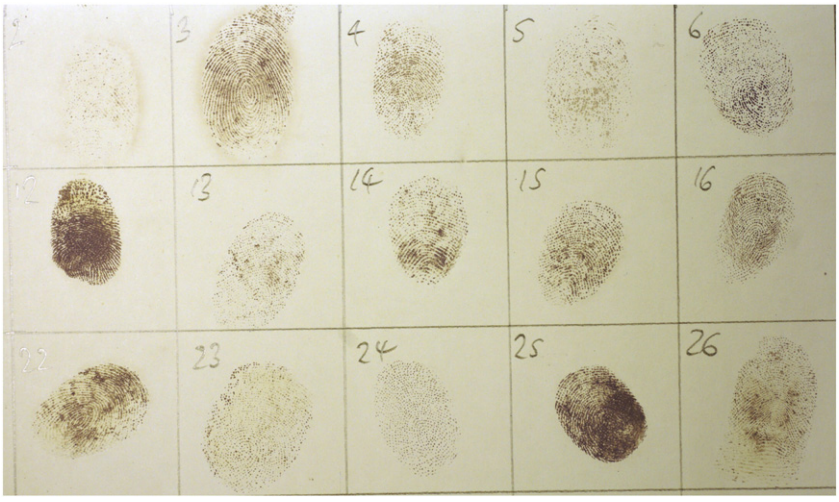


Fig. 9. An example of a 'multiple donor' study, showing marks from different donors developed using iodine fuming on semi-glossy paper.

2.2.2. Depletion series

The concept of a depletion series is used in the determination of the relative sensitivity of a technique. A series of finger marks is deposited by successive contacts of the same finger with the surface. In theory, each contact will leave progressively less residue on the surface and therefore the further down the depletion series that the development technique continues to develop finger marks, the more sensitive it is (although in practice this is only true if each mark is deposited with a consistent pressure) [16].

The number of finger marks that should be deposited in a depletion series will depend on the type of surface being used and the constituents that the process targets. In general, on a porous surface the residues are rapidly absorbed by progressive contacts and by the sixth finger mark very little material remains to be deposited, therefore the number of marks in the depletion series need not exceed this by much.

On non-porous surfaces many candidate techniques may continue to develop full ridge detail beyond the tenth mark in a depletion series and twenty or more marks may be required to begin to see the differences in sensitivity. In the case of finger marks on adhesive tapes, some techniques may still develop marks after the fiftieth contact. It may therefore be necessary to carry out a pre-experiment to establish the range at which differences begin to be seen between techniques and select the region of the depletion series which is to be developed accordingly. If it is known that both techniques being compared will consistently develop marks 1–20 for most donors, then there is no need to develop these in the experiment and the experiment should focus on marks 20–30 instead.

2.2.3. Split fingerprints

Because of the inherent variability in finger mark composition between donors and also for finger marks from the same donor, whenever possible efforts should be made to minimise the potential impact of this variability on interpretation of the results. One of the ways of achieving this is by the use of split finger marks, where a depletion series of finger marks is deposited and physically divided in two along the centre line. Each half of the depletion series is then processed with a different development technique, and then recombined for an assessment of the number and quality of the developed marks, Fig. 10. Wherever possible this should be the test method of choice.

The advantage of this method is that it is possible to directly compare half finger marks with the same chemical composition, quantity of material and pressure at time of donation, thus eliminating many of the uncontrollable variables from the comparison. A practical example of a split depletion experiment is illustrated in Fig. 11.

Split fingerprint experiments are best suited to surfaces such as paper which can be easily split after fingerprint deposition. There are many surfaces (e.g. glass) that are not easy to split, and there are also development processes for which the presence of a split edge may create edge effects which are not representative of how

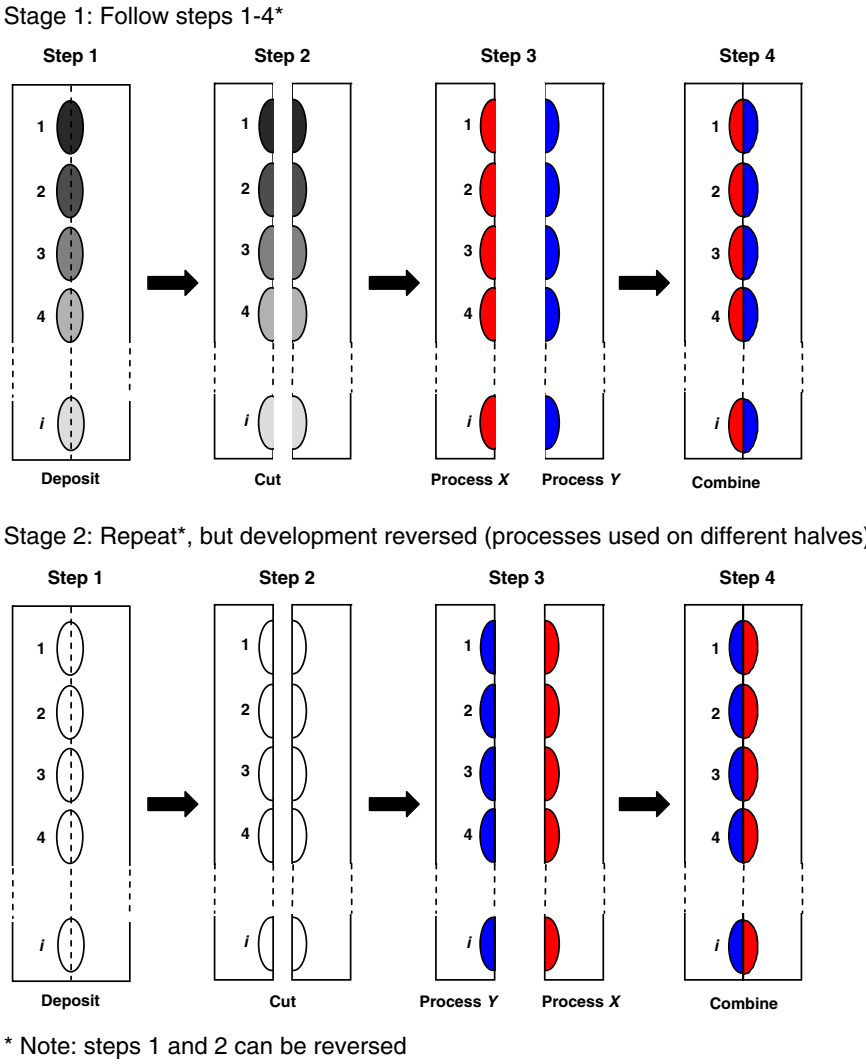


Fig. 10. Schematic diagram showing the split depletion method.

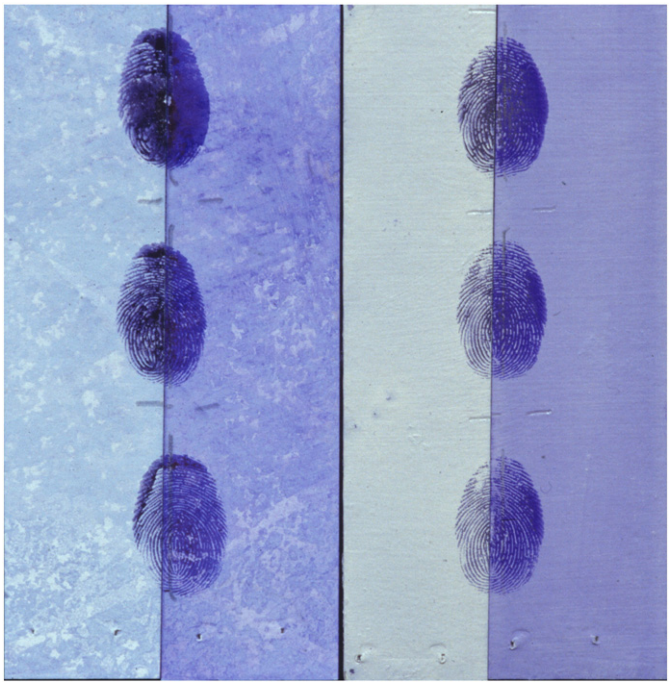


Fig. 11. Split depletion series between two acid dyes being evaluated for enhancement of marks in blood, tested on both wallpaper and a painted wall. Results show similarity in sensitivity and colour intensity of developed marks, but formulation on the right shows greater background staining.

the process would work on a continuous surface (e.g. powdering). For these situations alternative methods need to be considered, examples being 'alternate depletions' and 'multiple depletions'.

2.2.4. Alternate depletions

Even the act of rubbing hands together before deposition will not ensure consistency of finger mark composition across all the digits on a hand. For this reason, the next best alternative to using a split finger mark is to use a depletion series deposited from a single finger across two separate surfaces, each of which can then be developed using a different process. During deposition of the depletion series, alternate finger marks are deposited between the two surfaces. For example, fingerprints 1, 3, 5, 7, 9 in the depletion series will be deposited on surface 1, and fingerprints 2, 4, 6, 8 and 10 on surface 2. Because this will slightly favour the process being applied to surface 1, the experiment should be repeated with the processes used for each surface reversed, Fig. 12.

The advantages of the method are that although not directly comparing the same finger mark, this test allows for a comparison of marks from the same depletion series. These finger marks will have similar chemical composition, quality and pressure, although this is not as tightly controlled as in a split depletion series. The method is also suitable for all surfaces.

The method is not suitable if comparing more than 3 variables or techniques because to the limited number of useful finger marks that will be deposited across the 3 substrates required.

2.2.5. Multiple depletions

The alternate depletion approach above can add complexity to an experiment, and another option is to use depletion series deposited by different fingers on different surfaces, Figs. 13 and 14. Although this means that there will be more variability between the finger marks deposited on each surface, this can ultimately be addressed by including more repeat experiments and extending the range of donors used so that the overall trends in enhancement technique effectiveness can be seen over any local variations in finger mark composition.

The advantages of the method are that it is suitable for all surfaces and processes and can be used to test 4 or more variables or techniques. However, more data will be required to draw conclusions due to the greater variability introduced by using different depletion series.

2.2.6. Range of donors

Laboratory trials on finger mark enhancement techniques should include finger marks collected from a range of donors that are as representative of a cross section of the population as possible. Trials should use a minimum of five donors, where possible selecting donors that usually give a range of responses to the development technique under test, Fig. 14. A minimum of five donors is recommended because this number provides sufficient flexibility to incorporate variations in donor age, sex and other factors into the experiment. A preassessment can be carried out using a large range of potential donors to establish which typically develop marks of good, medium and poor quality when treated with the technique under test. The experiment should then utilise a mix of donors, for example 2 good, 2 medium and 2 poor donors in a pool of 6.

If the donor quality is unknown, a mix of donor ages and donors of both sexes should be used instead. The occupation of the donor can also have an effect, for example manual workers may have more worn, drier hands than office workers. Health (physical and mental), diet and medication will all directly affect the balance of chemicals within the body and hence the composition of sweat from the various glands. If the effects of any of these particular factors on fingerprint development are being assessed, it may be necessary to select donors on more rigorous criteria such as smoker vs. non-smoker, or vegan vs. meat eater.

The paragraphs above do not proscribe the number and type of donors that must be used, they give indications of some of the factors that may need to be taken into account when designing an experiment and the final selection of donors will depend on what is being evaluated. In general, as many different donors as is practicable should be used.

2.2.7. Contaminants

In some experiments, the technique being evaluated is intended for use in developing fingerprints in contaminant, examples being acid black 1 for developing finger marks in blood and solvent black 3 for developing finger marks in grease. In these cases it is the contaminant that is important and it is not necessary to use a range of donors. A finger from a single donor is dipped in the contaminant and a depletion series laid (which may consist of in excess of 20 contacts), with the aim of seeing how far down the depletion series the contaminant can still be detected. The use of a real finger is found to give more representative results because fake fingers do not interact with contaminant in the same way and do not deform realistically when applied to a surface.

2.2.8. Age of mark

Research [6] has established that finger mark composition changes with time after deposition, due to mechanisms such as loss of moisture and other volatile compounds, chemical reactions and oxidation. The extent to which this occurs is influenced by several factors, including surface type and condition, level of exposure to direct sunlight, ambient temperature, humidity, and air flow across the surface. It is therefore not valid to test enhancement techniques on freshly deposited finger marks alone, and the experimental programme should include finger marks across the range of ages that are likely to be encountered in an operational laboratory.

Finger marks at a scene attended by a crime scene investigator may be powdered within hours of deposition, whereas some articles

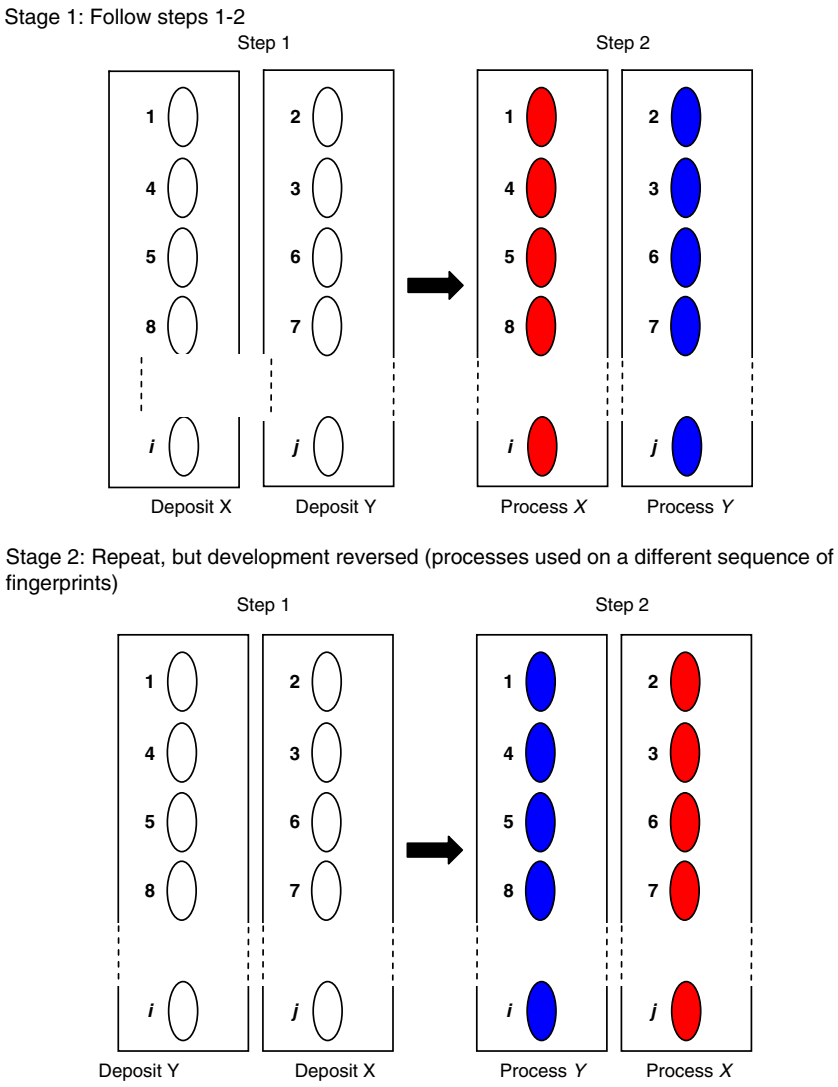


Fig. 12. Schematic diagram showing alternate depletion method.

related to a crime may not be recovered for several weeks and remain in storage for even longer prior to treatment. In practice the most significant compositional changes occur within the first few days, and therefore a typical experiment may include finger marks 1 day, 1 week and 1 month old.

Ideally, a technique will work equally well regardless of how old a finger mark is. However, even if the results show that a technique will only work well on recently deposited finger marks this knowledge is valuable in itself because it sets the constraints on when and where the technique can be applied.

2.2.9. Surfaces

It is extremely rare that a finger mark enhancement technique is only applicable to a single, clearly defined type of surface. The surface the finger mark is deposited onto may have a significant effect on the subsequent effectiveness of the development technique. Therefore the comparative experiments should be carried out using a range of surfaces that are representative of those that the technique may be applied to in operational work. Typically, five or six representative surfaces are selected, although the more that can be used the more robust the final conclusions will be. Some examples of groups of surfaces that have been used in evaluation of development techniques are given in Table 2 and shown in Fig. 15.

The surfaces used in these experiments are generally cut to an easily handled size, with non-porous surfaces being cleaned prior to finger mark deposition using soapy water and then ethanol so that most types of pre-existing contaminant will be removed.

2.2.10. Environment

If the technique under evaluation is intended for use in particular environments, it will be necessary to include an assessment of the effect of that environment in the trial. For example, if an evaluation is being carried out of a technique intended for use on wetted surfaces, the surfaces should be subjected to conditions that replicate wetting [17]. If techniques capable of developing finger marks on surfaces exposed to high temperatures are required, the deposited fingerprints should be placed into temperature controlled ovens for set periods of time [18]. The methods used to simulate the required environment can be defined by the researcher, but should try to replicate the conditions most commonly encountered in operational work wherever possible.

2.2.11. Grading schemes

To enable comparisons of relative technique effectiveness to be carried out, it is necessary to have some means of converting the visual appearance of the developed mark to a numerical value. The vast majority of those conducting research into finger mark development

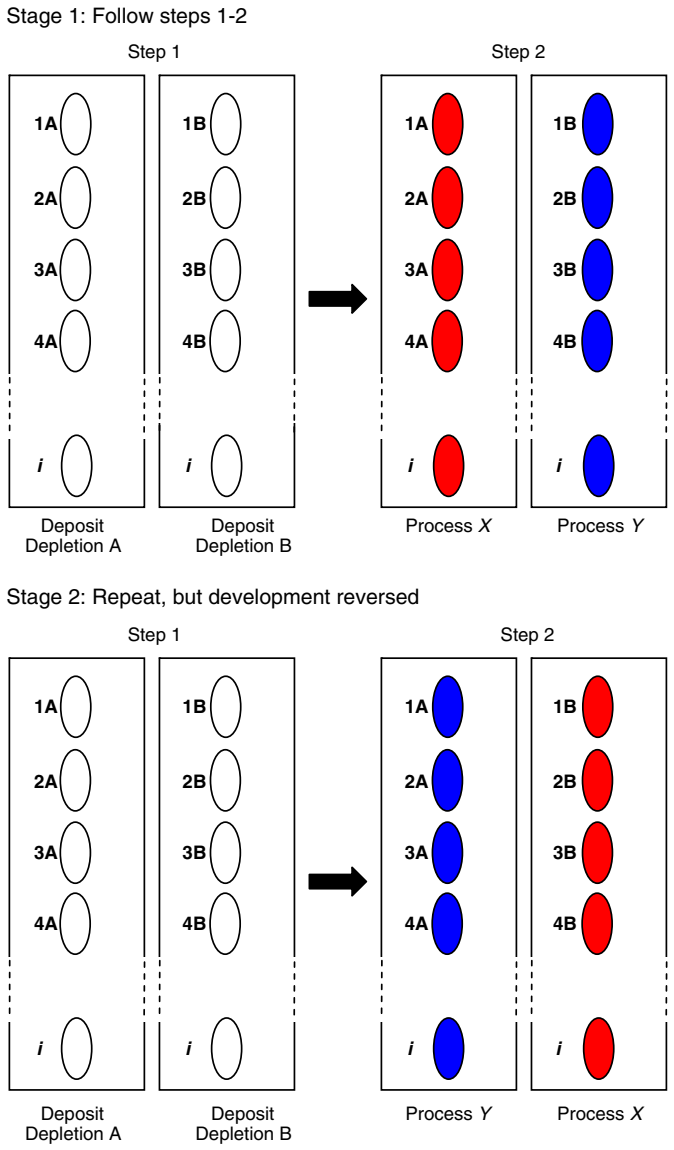


Fig. 13. Schematic diagram of the multiple depletion method.

are not identification experts, and therefore such a method should not rely solely on the ability to detect and count minutiae (features such as ridge endings and bifurcations) in the developed finger mark. However, the grading method used should bear some relation to the qualities of the finger mark that an identification expert would use, namely the extent and clarity of the ridge detail that has been developed. One such scoring system that has been used in various forms by the Home Office for over 35 years is outlined in Table 3 with examples of each grade illustrated in Fig. 16.

This grading system is subjective, and relies on all finger marks being scored by the same person. It is also necessary that the person carrying out the grading keeps a consistent approach to all finger mark assessment exercises. More objective ways of assessing and potentially scoring the quality of a developed mark are being researched, but none is currently fast enough to carry out a rapid assessment of the hundreds and potentially thousands of finger marks that may be developed during the course of a comparison experiment.

In some experiments other factors may need to be taken into consideration to give better differentiation between the techniques being evaluated. In such cases it may be necessary to apply secondary grades relating to factors such as contrast, background development, damage and ridge continuity, and taking the grading of both the primary and secondary attributes into account when assessing the results.

Although those most often carrying out this type of assessment are not identification experts, there is an approximate relationship between the grades outlined above and fingerprint identification. In general, marks of grades 3 and 4 will contain a sufficient area of clear ridge detail for an identification to be made. Marks of grade 2 could occasionally be classed as 'identifiable' if assessed by an expert, but this would depend on if the limited area of ridge detail developed contained sufficient minutiae, which may be the case if the developed portion included the core or a delta.

2.2.12. Interpretation

The finger mark grades obtained from the experiments outlined above can be used in several ways to enable comparisons to be made between the effectiveness of techniques in developing marks of identifiable quality. At the most basic level, the average grade recorded for marks developed by a particular technique across all surfaces, finger mark ages and donors gives some indication of how well a technique is performing. Counting the number of marks giving grades 3 or 4 ('identifiable' marks) also gives an indication of the technique's effectiveness. More complex analyses such as ANOVA may be used to identify which variables are having the most impact on technique effectiveness [18], and subsets of the data can be studied to focus on the effect of age alone, or see whether different trends are observed on different surfaces. Results obtained from separate experiments conducted at different times should be compared with caution as many of the variables (e.g. donor sets, temperature/time of year) may have changed.

The difficulty and complexity in applying statistical models to analyse the results from finger mark experiments should not be underestimated. This raises questions in relation to the number of deposited marks required to ensure sufficient robustness within the given experiments. The number of developed marks required to provide this robustness will vary with the technique or process' position in the sequence of techniques or processes used in a given set of experiments.

In devising the appropriate number of repetitions it is essential to understand the nature of the application of these techniques. In a sequence of enhancement processes for a given surface, those which are non-destructive in nature should be examined initially and require less indication of their effectiveness in enhancement of a deposited residue. This is because they will not have an adverse effect on the deposit or compromise any further processing. Experiments on processes that are potentially destructive to the mark will require significantly more marks to be developed both to establish their effectiveness over other competing processes and to confirm exactly how destructive they are so that their position in the sequence of development processes may be determined. However, the most destructive (and therefore last to be utilised) process requires fewer marks to be developed simply because no other process is used afterwards.

Therefore, the number of finger marks required to confirm a level of confidence in the effectiveness of a specific enhancement process would resemble a Gaussian distribution with most verification required for those in the middle of the process sequence. Thus, as this suggests, the exact number of finger marks deposited will relate specifically to the research objectives of a particular experiment or set of experiments as well as the nature of detection methods and surfaces being employed.

2.2.13. A typical experiment

A typical experiment to carry out an initial comparative assessment of a technique for a porous surface would have 5–6 donors, 5–6 surface types, 5–6 depletions, and 2 ages of finger mark, which means 250–432 finger marks which are split in half and each treated

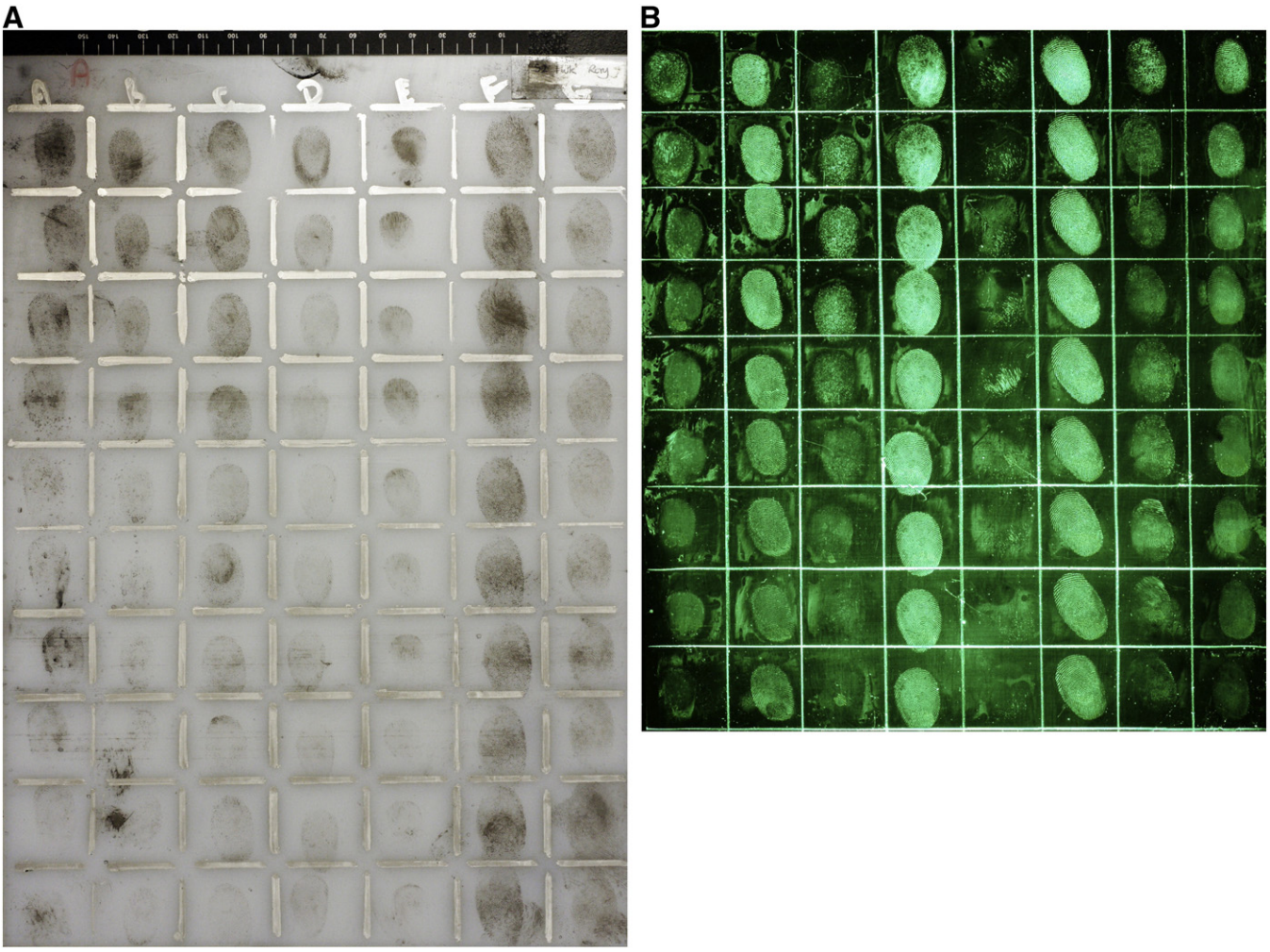


Fig. 14. Examples of multiple depletion series, a) black powder suspension on white polythene sheet, b) cyanoacrylate fuming/basic yellow 40 dye on ceramic tile.

with a different formulation. To get a comparison set of 3 variables or techniques would require 3 experiments (A vs B, B vs C and C vs A). However, to draw meaningful conclusions the set of 3 experiments should be carried out at the simultaneously using the same donors.

It would be expected to repeat each comparison set at least 3 times to give a fair indication of performance. However, if a variation or technique were clearly underperforming it would be possible to drop that from the trial.

This experiment size is scalable depending on the purpose of the experiment, as described above.

Table 2
Examples of the types of surface used for finger mark deposition within a generic material 'class'.

Generic description	Surfaces used
Smooth, non-porous 'indoor' surfaces	Glass, smooth melamine, ceramic tile, gloss painted wood, uPVC
Porous surfaces	Printer paper, writing paper, lined notebook paper, brown envelope, white envelope, newspaper, magazine
Adhesive tapes	Parcel tape, 'duck' tape, clear tape, electrical tape, masking tape
Plastic bags	Carrier bag, 'waxy' carrier bag, bin bag, crisp packet, sandwich bag, plastic courier envelope
Wall coverings	Paper wallpaper, vinyl wallpaper, matt emulsion paint, satin paint, silk paint, gloss paint

The objective of this phase of testing is to establish whether any new technique is more or less effective than current processes and merits further study. Tests may also be required to obtain some idea of where the process could fit into a sequential treatment regime (see below), and to assess its compatibility with recovery of other forensic evidence types.



Fig. 15. Examples of the different types of adhesive tapes that have been used in laboratory trials.

Table 3
Outline grading scheme used for assessment of developed marks.

Score	Level of detail
0	No evidence of mark
1	Weak development; evidence of contact but no ridge details
2	Limited development; about 1/3 of ridge details are present but probably cannot be used for identification purposes
3	Strong development; between 1/3 and 2/3 of ridge details; identifiable finger mark
4	Very strong development; full ridge details; identifiable finger mark

2.3. Establishing finger mark development sequences

In order to establish finger mark development sequences, some means of testing the overall effectiveness of the sequence, the individual effectiveness of the techniques within it and the impact of each process on subsequent techniques is required. Some of the methods that can be used for these purposes are simple adaptations of the methods used to directly compare the effectiveness of individual techniques in laboratory trials, and these are outlined below.

2.3.1. Sequential processing of multiple donor panels

The 'multiple donor' approach can be adapted to the evaluation of sequential development routines. In such an experiment a single finger mark is collected from many different donors, trying to obtain as representative a population of donors as possible. All of these finger marks are deposited on a single surface, which is then exposed to the desired environment for the selected period of time. The surface is then exposed to the first development technique, and the number and quality of the finger marks that are developed by that technique is recorded.

The surface is then exposed to the next development technique in the sequence under test, and the grading exercise is repeated. This process is repeated for all of the techniques in the proposed sequence in turn, Fig. 17. This should be repeated over a range of different surfaces and for marks of different ages. If two candidate sequences are being compared, the surfaces to be used for each test sequence should be prepared at the same time using the same donors.

The overall effectiveness of the sequence is evaluated by assessing the number of the originally deposited finger marks that are developed to a quality sufficient for identification. If a finger mark is developed by more than one of the techniques in the sequence, it will only count as a single developed fingerprint. The effectiveness of each individual technique can be determined by recording the number of marks that are developed to an improved quality over the previous technique, the number of additional marks it develops over the previous technique, and the number of unique marks it develops as part of the overall sequence. Such an analysis may enable the removal of a

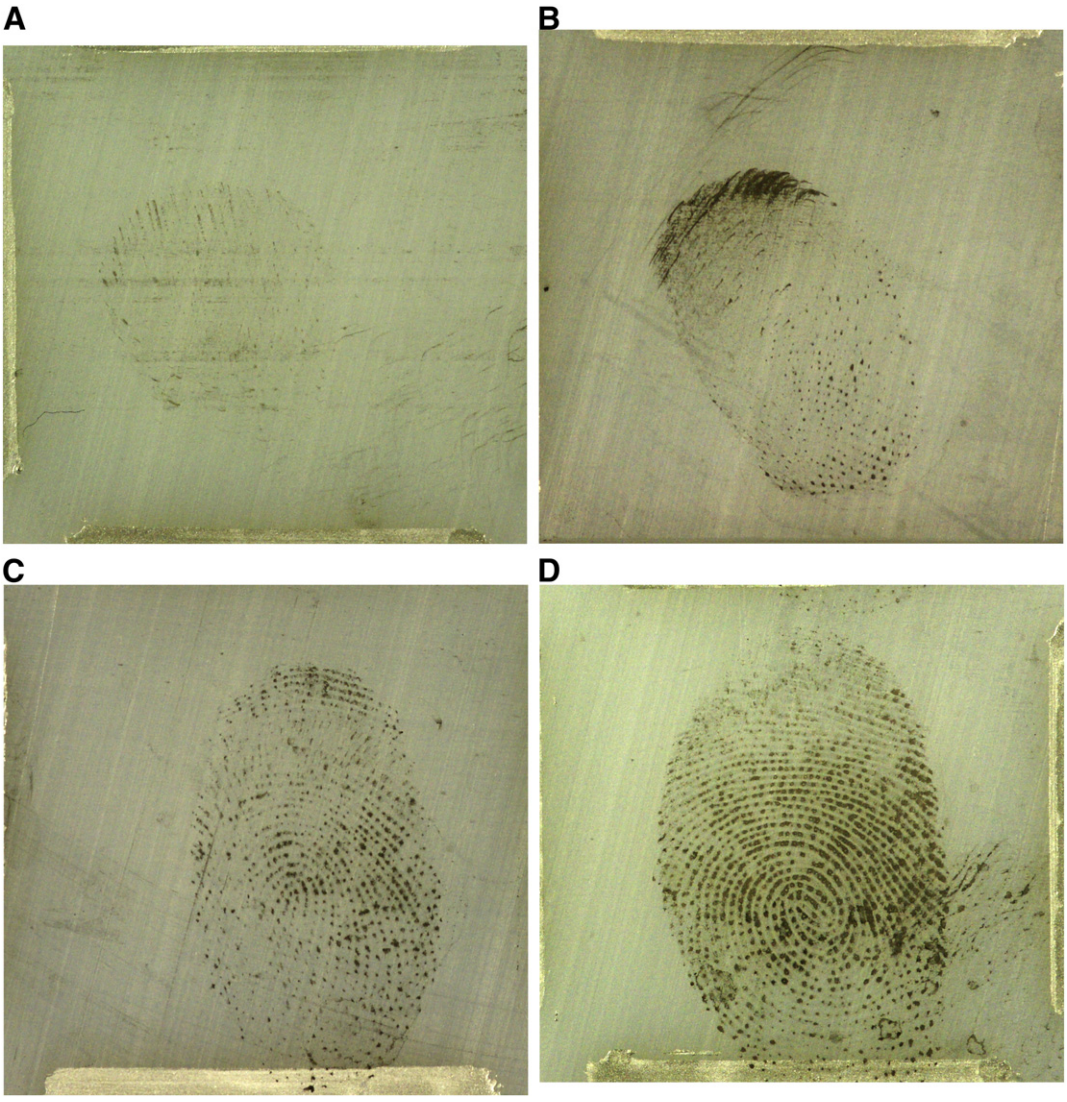


Fig. 16. Developed finger marks representative of grades 1, 2, 3 and 4 (developed using black powder suspension on polyethylene sheet).

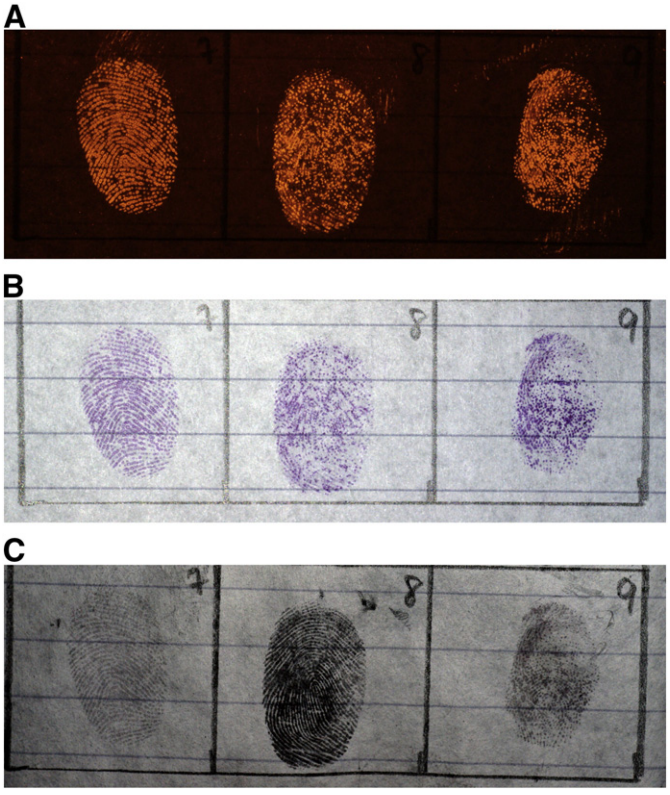


Fig. 17. An example of a series of finger marks on notebook paper exposed to a sequence of processes, A) DFO, B) ninhydrin and C) physical developer.

technique from a sequence if it is demonstrated that it gives little benefit and develops no unique marks. Optimum sequences are those that develop the highest proportion of high quality finger marks from the number originally deposited.

Fig. 18 shows typical results obtained from a sequence of three processes being tested across a range of paper types. Tests were performed for marks of different ages, recording both the number of donors out of 36 that gave some ridge detail (grades 1–4) and identifiable marks (grades 3–4). These results could then be compared to those from an alternative sequence to establish which sequence typically developed marks across the broadest range of donors.

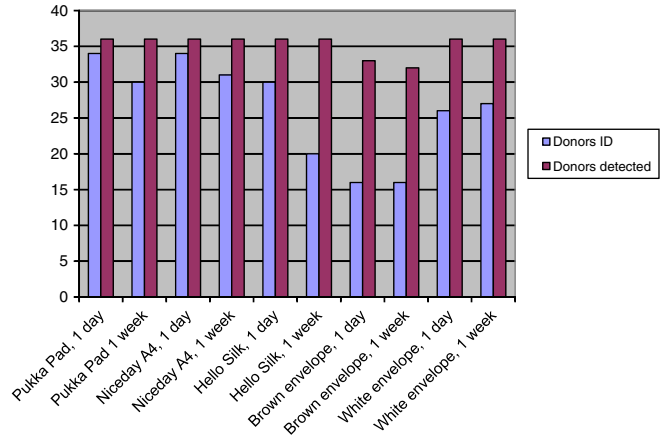


Fig. 18. Effectiveness of a test sequence evaluated using a multiple donor study, results indicate number of marks detected and 'identifiable' for a set of 36 donors.

2.3.2. Sequential processing of split finger marks

The potential impact of one enhancement process on the subsequent application of another can be investigated by the 'split finger mark' method. This is particularly important where it is suspected that there is an incompatibility between two of the techniques that are being proposed for use in the sequence, or that the optimum order in which two techniques are to be used needs to be determined.

In this experiment, a depletion series of finger marks are deposited down a centre line, either on a surface that is capable of being subsequently split (such as paper), or has been pre-split. The surface is the divided and each half exposed to a different treatment or sequence of treatments. In an example where techniques A and B are being proposed for use in the sequence A–B, one half of the finger mark is treated using technique B (the 'control' to establish how many finger marks technique B would develop), and the other treated by technique A followed by technique B (the test sequence). The two treated halves of the finger mark can then be recombined and the effect of technique A on technique B determined, Fig. 19.

By carrying out a series of these tests it is possible to determine which processes are incompatible and which can be used in sequence without detrimental effects.

3. Pseudo-operational trials

The principle of the pseudo-operational trial is to establish whether the results obtained in laboratory trials are replicated on articles/surfaces typical of those that may be submitted to a fingerprint laboratory, or to distinguish between closely equivalent formulations that cannot be separated in laboratory trials. Some processes, like ninhydrin, do not always give results equivalent to those seen on operational work during trials using planted finger marks, and therefore it is necessary to evaluate such processes on more operationally representative material.

The material for a pseudo-operational trial is obtained by collecting representative surfaces from as many different sources as possible, some examples of materials used in pseudo-operational trials being summarised in Table 4 and illustrated in Fig. 20.

Once collected, these articles should be divided into batches which are as closely equivalent in the number, type and source of articles as possible. Each batch should then be treated with a separate technique. The items should be recovered, handled and processed as if they were from a criminal case, with no special handling to deposit additional marks. Methods that have been used in such trials include the use of batches of fraudulently passed cheques, divided into batches containing equivalent numbers of cheques from the same banks, and where possible from the same account name. Larger items such as plastic bags may be cut top to bottom into halves or quarters and then each piece can be treated with a different process or sequence of processes. The number of articles being treated needs to be sufficiently large for trends to be seen. Previously reported pseudo-operational trials have utilised batches of 75 fraudulently passed cheques [19–21] or 100 plastic bags to get enough information for this phase.

For the most part, the items collected will have an unknown history, which more closely represents the situation in an operational laboratory. In addition, the finger marks that are present on the items will not represent the full, controlled contact that has been used to apply planted fingerprints in laboratory trials. It is more likely that the finger marks developed will be partial contact marks, or will exhibit smudging or movement. For this reason, it is not possible to use the grading system described above and some means of determining whether a developed mark is of identifiable quality or not is required. The best way of doing this is to obtain the opinion of a finger mark identification specialist. If this is not possible, the researcher can take some basic training in identification and assess each developed mark themself, rating those with more than 8 (or any other arbitrary

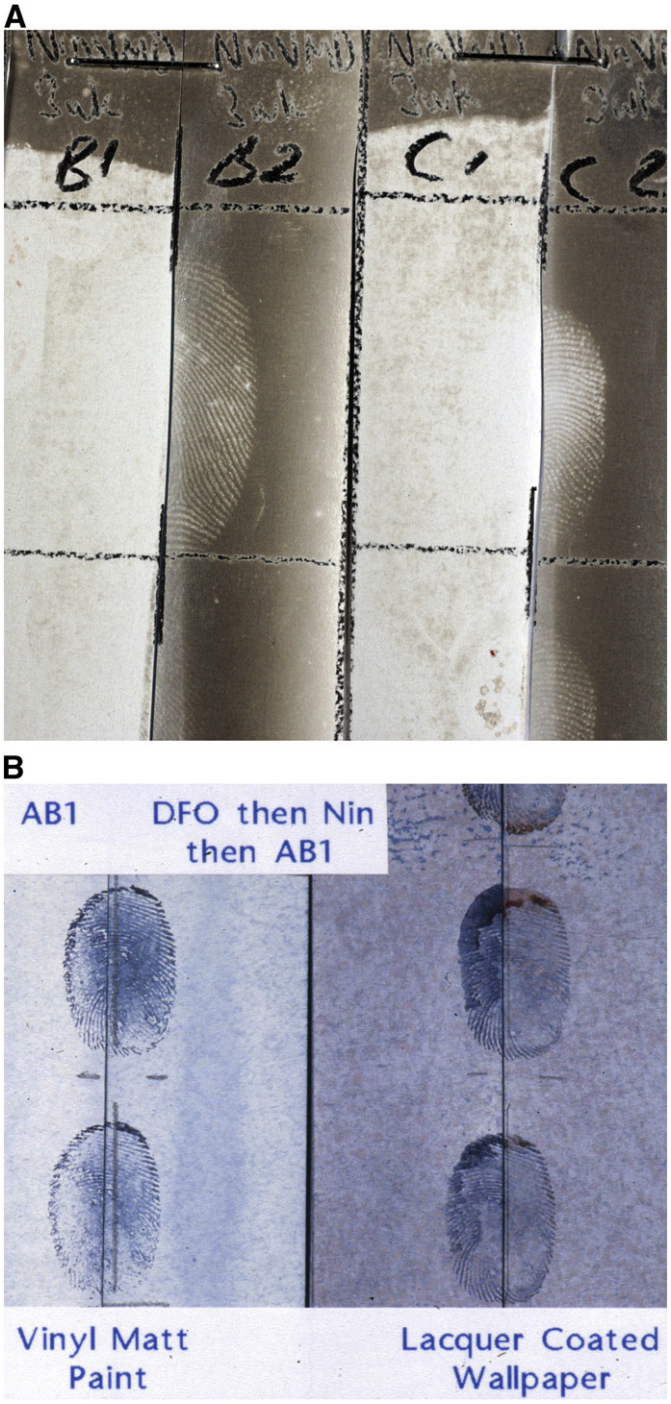


Fig. 19. Split deletion series used to examine compatibility of processes in sequential treatment routines, A) the detrimental effect of ninhydrin when used before vacuum metal deposition on a semi-porous surface, and B) an illustration that DFO and ninhydrin have no detrimental effect on subsequent use of acid black 1 when used to detect blood on walls.

value pre-determined in discussions with identification specialists) visible minutiae as identifiable. This assumes that if the non-specialist can clearly see 8 points, an expert will be able to locate several more. An alternative method that does not rely on any subjective assessment is to measure the area of clearly defined ridge detail developed by the technique.

A grading scheme more representative of operational work has been devised for pseudo-operational trials. This scheme takes into account the area of ridge detail present, the continuity of the ridges and the contrast with the background. The area of ridge detail relating to each grade was developed in consultation with finger mark identification experts and related to 'second level detail' point counts. A variety of areas were examined on developed finger and palm marks and it was found that almost all areas over 64 mm^2 contained at least 8 'second level detail' points which most UK police forces consider sufficient for initiating a database search.

Table 4
Examples of the types of article that have been used in pseudo-operational trials.

Generic description	Surfaces used
Non-porous surfaces	Drinks cans, sandwich containers, carrier bags, glass and plastic bottles
Porous surfaces	Newspapers, envelopes, printed documents, notebook pages
Thermal papers	Till receipts, betting slips, ATM receipts, fax paper
Contaminated surfaces	Take away food packaging, drinks cans, disposable plates and cutlery

3.1. Area grading scheme

1 Areas of development between 1 and 63 mm^2
2 Areas of development over 64 mm^2

3.2. Sub categories within grade 2

a Continuous ridges, good contrast
b Continuous ridges, poor contrast
c Discontinuous ridges, good contrast
d Discontinuous ridges, poor contrast

The continuity measure may not be always suitable as techniques targeting eccrine constituents are more likely to produce finger marks with discontinuous ridges.

The scores from the experiments can be interpreted in a number of different ways. The easiest way to interpret the results is to count the total number of finger marks developed. However, sometimes articles are encountered on which many times the average amount of finger marks are developed and these may bias results in favour of a particular technique. It is possible to discount results from these 'outlier' articles if there are only a few but if a large variation in the number finger marks developed between different articles is commonly seen then this becomes difficult.

Another method of analysis is to count the number of articles or mock cases (where possible) on which finger marks have been



Fig. 20. Examples of articles used in pseudo-operational trials on plastic packaging materials.

developed. This can be a more subtle way of measuring effectiveness but also has its disadvantages, for example where both processes develop marks on a similar number of cases or articles, but one process consistently develops more finger marks overall. Ideally, both methods of analysis should be performed and the trends from both compared.

The overall objective of this phase is to build confidence that the improved performance demonstrated in laboratory trials can be replicated on exhibits representative of real casework and to minimise the risk that evidence may be lost during any subsequent operational trial.

4. Full operational trial

In this final phase of the work a trial is carried out on live casework, running the techniques under comparison alongside each other with the objective of establishing whether there is operational benefit in introducing the new technique in place of existing processes.

In the operational trial, cases should be randomly distributed between the techniques or formulations being used for the trial. This may involve some intelligent distribution so avoiding one process being used to treat excessive amounts of one type of surface. For instance cases involving cheques, or brown envelopes or newspapers etc. would be evenly distributed between processes. It is not generally worth splitting cases between processes as most cases passing through UK police laboratories involve fewer than 3 items of one type of surface.

An operational trial should last as long as it takes to establish the effectiveness of the techniques of formulations in the trial. This may require treating more than 100 cases with each competing technique or formulation to assess effectiveness; however the number required could be much fewer if confidence in the result is rapidly obtained.

If possible this phase should be carried out blind so that the person carrying out the marking up process is unaware of which technique has been used on which article.

Finger marks developed in this phase should be assessed by an identification specialist using normal casework standards, counting finger marks that can be identified and the number of cases and articles marks are developed on [22,23].

Results can be interpreted in the same ways as with the pseudo-operational trial. However, because operational work is case-based and that cases a single process is used on each case, analysis by positive cases has more meaning. The process that develops finger marks on most cases will effectively be developing a wider range of finger mark types.

It is essential that the number of identifiable marks that are being developed by each technique is continuously monitored. If it appears that the new technique is consistently underperforming and there is a risk that evidence is being compromised, then that technique should be removed from the trial. Any 'outliers' should also be removed from the analysis of the data, because overall technique effectiveness may be skewed by a single exhibit with a particularly high (or unusually low) number of fingerprints on it.

If the new process demonstrates operational benefits over a prolonged period (typically several months) there will be a case for introducing it into operational use and work will then proceed to preparation of standard operating procedures.

5. Conclusion

A methodology that describes four phases of validation for finger mark enhancement techniques has been outlined. This methodology represents the approach taken by the UK Home Office during the work to select the processes and sequences outlined in the Manual of Fingerprint Development Techniques.

This paper does not discuss how methods are made practical and safe for appropriate use. Whatever is required in this respect should be carried before operational trials are conducted as this may bias officers carrying out the trial against the new method if it is too difficult to use, and performance may suffer as a result. Other points to consider in addition to effectiveness prior to final implementation include: cost (e.g. of chemicals and equipment), time (and human resources) taken to develop marks, complexity of technique, and effect on other types of forensic evidence.

Techniques and processes do not need to go through each phase of development if it is considered appropriate for one or more to be missed out, although this is unusual. For example the protein stains for blood did not have formal pseudo-operational or operational trials as once the effectiveness of these dyes has been proved on deliberately deposited finger marks contaminated with blood[14, 24, 25] they will work the same on operational work because a known contaminant is being enhanced in both cases. On the other hand, the CFC-free ninhydrin process [23] went through all four phases of development before being implemented.

It should be noted that the results obtained by following the validation methodology described above may only be valid for the locality that the tests are performed in. There have been observations from researchers worldwide that performance of some processes appears to vary from one country or region to another. Differences in climate, the types of materials or even differences in diet encountered worldwide may mean that different conclusions are obtained from nominally similar comparative studies. However, if all researchers follow the guidelines outlined in this methodology it would bring some consistency to the way in which finger mark enhancement processes are developed and reported, and make it easier for law enforcement agencies to identify the most effective processes.

Acknowledgements

The authors would like to thank Dr Niamh Nic Daeid of Strathclyde University for reading an early version of this manuscript, and for her helpful comments that have enabled preparation of the current paper.

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